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Evidence-Based Practice Approach on Protocol Development on Prevention of Hypoglycemia for Patients with Hypertriglyceridemia-Induced Pancreatitis

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Introduction/ Background

For the past five years, the Rapid Response Team in Henry Ford Providence Novi has witnessed :

- An increase in the incidence rate of rapid response activation for hypoglycemia in patients receiving insulin infusion as a treatment for hypertriglyceridemia-induced pancreatitis in our acuity-adaptable nursing units.
- There is currently no order set/protocol for the use of insulin infusion for hypertriglyceridemia-induced pancreatitis (HTGP). Patients with severe hypertriglyceridemia (HT), defined as:
  - a triglyceride level greater than 1000 mg/dL,
  - are at risk for the development of acute pancreatitis, premature atherosclerosis, or cardiovascular disease.

The increase in hypoglycemia reaction and the delay in decreasing triglyceride levels resulted in:

- longer Length of Stay (LOS) with an average of 7-8 days or more .

Insulin is a high-alert medication, and an order set guiding safe, evidence-based therapy is warranted.

Purpose/Aims

- Severe hypertriglyceridemia requiring hospitalization for intravenous insulin to lower triglycerides and prevent complications of pancreatitis is becoming an increasing problem with little consensus treatment evidence.
- A literature review was made to evaluate the latest Evidence-Based Practice recommendation that will guide a multidisciplinary team to formulate a treatment protocol for HTGP that will prevent an occurrence of hypoglycemia as a sequela of using insulin infusion and shortened treatment to achieve triglyceride level 500 Mg/dl.

Methodology

- The protocol for preventing hypoglycemia for patients with HTGP uses a conceptual **framework** of ***Knowledge Translation***.
  - Strauss et al. (2009) emphasize the recognition of gaps in translating knowledge to action, will lead to change behavior, practices and policy.
  - The multidisciplinary team comprises a clinical pharmacist, critical care nurses, laboratory, critical care nursing director, bed management, personnel intensivists, endocrinologists, and internal medicine physicians believe in the complexities of knowledge translation and how effectively integrating knowledge into practice involves an embodied process of inquiry and action.

Methodology (continued)

The multidisciplinary team formulated the protocol based on the literature review and internal networking with other Henry Ford Providence Ministries:

- Yang et al. (2024) emphasize the importance of prompt recognition of hypertriglyceridemia in the setting of acute pancreatitis is essential in both the initial and long-term management of this disease and are essential to prevent recurrent acute pancreatitis.
- Furthermore, Gang et al. recommend insulin infusion at a rate of 0.1 to 0.3 U/kg per hour to lower TGs level irrespective of concurrent hyperglycemia.
- Frequent blood sugar checks every 1 hour should be done to prevent hypoglycemia and if present, dextrose solution should be added to infusion.

Before the creation of this protocol, physicians and nurses did not have a treatment protocol for HTGP patients. These patients were most likely admitted to acuity-adaptable/step-down nursing units.

- The nurse-to-patient ratio in these nursing units is usually 1:3 or 1:4, resulting in missed blood glucose monitoring every hour and late recognition of hypoglycemia.
- The formulated protocol is a multidisciplinary approach and can be implemented in the ICU so that patients can have closer monitoring, thereby preventing the occurrence of hypoglycemia. The Drug Special Panel (DSP) Critical Care has approved the implementation of the protocol.

HGTP Order Set

ICU Insulin Drip for Severe Hypertriglyceridemia (> 1,000 mg/dL)

Diet

☑ Diet NPO
Start Date T;N
Comments: Until serum triglycerides are less than 500 mg/dL.

Nursing Orders

☑ Blood Glucose POC
q1hr

☑ Notify Provider
For symptomatic hypoglycemia and/or blood glucose less than 70 mg/dL

☑ Notify Provider
For symptomatic hypoglycemia and/or blood glucose less than 70 mg/dL. Notify the provide for glucose less than 200 mg/dL to initiate dextrose-containing fluids if not already ordered

Continuous Infusions

Review and discontinue all other diabetes medications (NOTE)*
Severe hypertriglyceridemia is defined as serum triglycerides greater than 1000 mg/dl (NOTE)*
If a patient presents with concomitant Diabetic Ketoacidosis (DKA) or Hyperosmolar Hyperglycemic Syndrome (HHS), please use ICU DKA/ HHS Power Plan instead. Do not transition insulin drip to basal/bolus insulin until serum triglycerides are less than 500 mg/dL (NOTE)*
Stop insulin infusion if blood glucose is less than 70 mg/dL. Treat per Non-Pregnant Adult Hypoglycemia protocol. (NOTE)*

☑ Insulin Oral Hypoglycemic Non-pregnant (SUB)*

Insulin Infusion - Select one

☑ Communication Order to Nursing
Start Date T; N, do not start insulin infusion if blood glucose is less than or equal to 119 mg/dL. Start dextrose D10% infusion at 125 ml/hr and begin insulin infusion when blood glucose is greater than 119 mg/dL.

☑ Communication Order to Nursing
Start Date T; N, do not start insulin infusion if potassium is less than 3.3. If the potassium level is less than 3.3, replace potassium first

Insulin Drip: Initial rate 0.1 unit/kg/hr. **Do Not Titrate to Blood Glucose (NOTE)***

☐ insulin lispro IV Infusion 100 units / 100 mL (IVS)*
Sodium Chloride 0.9% diluent
100 mL, 0.1 unit/kg/hr.
Comments: Do Not Titrate to blood glucose.
insulin lispro for additive
100 unit(s), 0.1 unit/kg/hr, Every Bag

For Hypertriglyceridemia. If at least a 25% reduction in triglycerides is not achieved after 24 hours of insulin infusion, you may consider increasing the insulin infusion rate (for example: increase from 0.1 unit/kg/hour to 0.2 unit/kg/hour. Stop when triglyceride level is less than 500 mg/dL. (NOTE)*

Insulin Drip: Initial rate 0.05 unit/kg/hr. Do Not Titrate to Blood Glucose (For patients with End Stage Renal Disease (ESRD) or Type 1 Diabetes (NOTE)*

☐ insulin lispro IV Infusion 100 units / 100 mL (IVS)*
Sodium Chloride 0.9% diluent
100 mL, 0.05 unit/kg/hr.
Comments: Do Not Titrate to blood glucose.
insulin lispro for additive
100 unit(s), 0.05 unit/kg/hr., Every Bag

For Hypertriglyceridemia. If at least a 25% reduction in triglycerides is not achieved after 24 hours of insulin infusion, you may consider increasing the rate of insulin infusion (for example: increase from 0.1 unit/kg/hour to 0.2 unit/kg/hour). Stop when triglyceride level is less than 500 mg/dL. (NOTE)*

IV Solutions

☐ Dextrose 10% (Hypoglycemia)
1,000 mL, IV Infusion, Infuse Over 125 mL/hr. (DEF)*
1,000 mL, IV Infusion, Infuse Over 150 mL/hr.
1,000 mL, IV Infusion, Infuse Over 200 mL/hr.

☐ Dextrose 5% with 0.9% NaCl + KCl 40 mEq
1,000 mL, 100 mL/hr (DEF)*
1,000 mL, 150 mL/hr
1,000 mL, 200 mL/hr

☐ Dextrose 5% with 0.45% NaCl + KCl 40 mEq
1,000 mL, 100 mL/hr (DEF)*
1,000 mL, 125 mL/hr

☐ Dextrose 5% in Lactated Ringers 1000 mL + 40 mEq KCl (IVS)*
Dextrose 5% in Lactated Ringers
1,000 mL, 100 mL/hr
KCL for additive

☐ Dextrose 5% in Lactated Ringers
1,000 mL, 125 mL/hr (DEF)*
1,000 mL, 100 mL/hr

☐ Dextrose 5% with 0.9% NaCl
1,000 mL, 125 mL/hr (DEF)*
1,000 mL, 100 mL/hr

☐ Dextrose 5% in Water
1,000 mL, 150 mL/hr (DEF)*
1,000 mL, 100 mL/hr
1,000 mL, 200 mL/hr

Medications

Start therapy when the patient is on an oral diet. Start T+1(NOTE)*

☐ fenofibrate
145 mg, PO (oral), q Day

☐ atorvastatin
10 mg, PO (oral), at bedtime

☐ atorvastatin
20 mg, PO (oral), at bedtime

☐ atorvastatin
40 mg, PO (oral), at bedtime

☐ atorvastatin
80 mg, PO (oral), at bedtime

☐ niacin 500 mg oral tablet, extended release
500 mg, ER Tablet, PO (oral), at bedtime

Laboratory

☑ Triglyceride
Spec type Blood, Collection priority Timed Study, Collection Dt/Tm T; N, Frequency q12hr

☑ BMP
Spec type Blood, Collection priority Stat, Collection Dt/Tm T;N (DEF)*
Spec type Blood, Collection priority Timed Study, Collection Dt/Tm T; N, Frequency q8hr

☑ Lipid Panel
Spec type Blood, Collection priority Stat, Collection Dt/Tm T; N

☐ Magnesium Level
Spec type Blood, Collection priority Stat, Collection Dt/Tm T; N (DEF)*
Spec type Blood, Collection priority AM, Collection Dt/Tm T; N, Frequency q8hr

☐ Phosphorus Level
Spec type Blood, Collection priority AM, Collection Dt/Tm T; N, Frequency q8hr

☑ Hemoglobin A1c
Spec type Blood, Collection priority Routine, Collection Dt/Tm T; N

Consults

☑ Consult Nutrition/Dietitian Adult
Please note insulin drip for severe hypertriglyceridemia only.

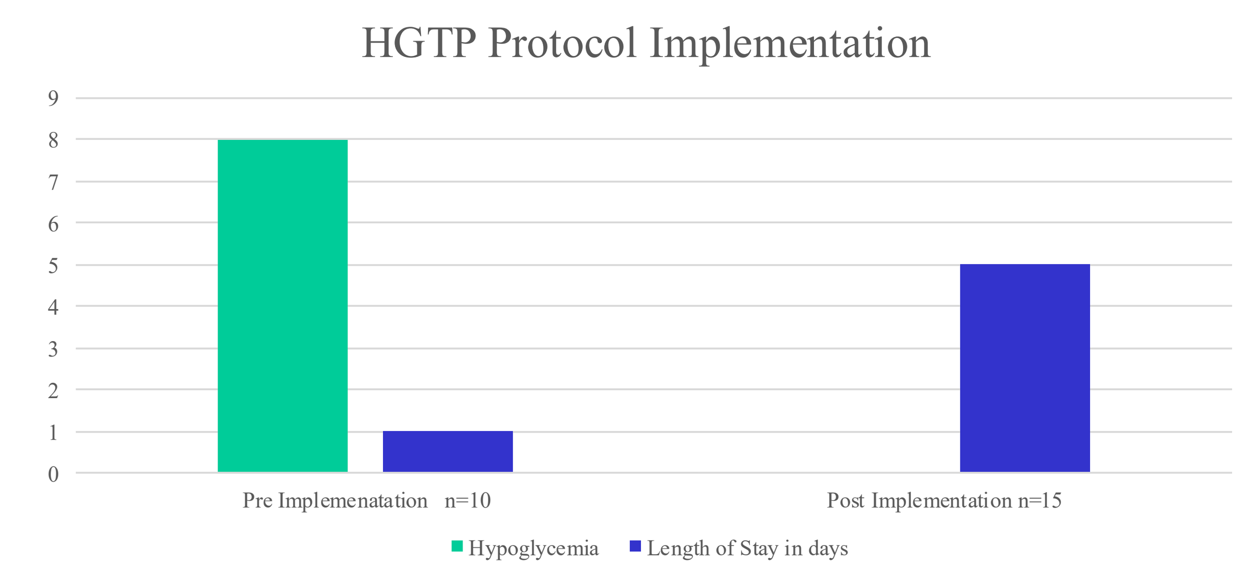
☑ Registered Dietitian to Manage Medical Nutrition Therapy
for Diet Education, Severe hypertriglyceridemia

Content last reviewed on 03/22/2024(NOTE)*

***Report Legend:**
DEF - This order sentence is the default for the selected order
GOAL - This component is a goal
IND - This component is an indicator
INT - This component is an intervention
IVS - This component is an IV Set
NOTE - This component is a note
Rx - This component is a prescription

Results

- Ever since the adaptation of the EBP to Protocol Development on the Prevention of Hypoglycemia for Patients with HTGP:
  - Hypoglycemia has not occurred, and
  - LOS has decreased to 4-5 days because of the triglyceride level less than 500 mg/dl. Achieved within 36-48 hours.



Conclusions

Evidence-Based Practice Approach on Protocol Development on Prevention of Hypoglycemia for Patients with Hypertriglyceridemia-induced pancreatitis showed :

- a decreased incidence of hypoglycemia,
- shorter treatment duration in achieving the triglyceride level < 500 Mg/dl,
- resulting in lesser hospital LOS.

References

- Gao, Lin; Li, Weiqin*. Hypertriglyceridemia and acute pancreatitis: clinical and basic research—a narrative review. Journal of Pancreatology 7(1):p 53-60, March 2024. | DOI: 10.1097/JP9.0000000000000153
- Garg, R., & Rustagi, T. (2018). Management of Hypertriglyceridemia Induced Acute Pancreatitis. *BioMed research international*, *2018*, 4721357. https://doi.org/10.1155/2018/4721357
- Straus, S. E., Tetroe, J., & Graham, I. (2009). Defining knowledge translation. *CMAJ : Canadian Medical Association journal = journal de l'Association medicale canadienne*, *181*(3-4), 165–168. https://doi.org/10.1503/cmaj.081229
- Request can be made for more of the references used.